

30 Minutes Expert

Harmonise that BIOS!

Changing BIOS settings isn't an act for the faint hearted. The BIOS gives users an avenue to change the most basic processes in a PC. It's also one of the most frequented stops when installing hardware. Unfortunately, the whole subject remains as obscure to many today, as it has always been. But with the right understanding, and possibly some tolerance on the side for good measure, you'll be able to tailor your computer to your way of working. Want to change the way your PC loads? Need to install a hard drive? Read on.

Getting through

In most PCs, particularly of the assembled variety, simply depress the [F1] or [Delete] key (a directive onscreen usually says which) a brief moment after you power up your computer. Usually, you have only a few seconds to take action, so make sure you're standing by.

■ The **Standard Options** menu offers the simplest level of setup for users. You can access basic settings such as the time and date, and even set up your floppy drive through this menu.

One of the most common problems faced by users has to do with fixed devices. This would include hard drives, CD and DVD drives or any other mass storage drive that is installed in a similar fashion. At times, when the computer's battery is weak or has been changed, the BIOS

settings revert back to the default settings, which may or may not comply with your specifications. As a result, hard disk error messages will arise. Drive models that date after 1997 should face no problems at all if you change the drive type to 'Auto'. In case you have a rather dated hard disk attached to the system, it's best to manually enter the drive's details such as sectors, cylinders and so on. You can usually find these settings listed on the hard disk itself.

■ In the **Advanced BIOS** category, you will be presented with a higher level of control. You can choose to have the booting process start from either a CD-ROM or alternative storage devices such as a Zip drive. Enabling anti-virus protection won't help you much with common viruses as it simply ensures that nothing can be written to the boot sector, a place where only the most dangerous of viruses encroach. With the Internet and its worms and Trojans, this basic level of protection does practically little.

■ **Power Management** helps your PC operate independently of you. There are a number of nifty things you can do by simply tweaking the settings in this area. Putting your computer off to 'sleep' and waking it up later is a clever principle for saving power and loading time. You can specify the exact time and date you want to power up or power down the system.

The key power saving modes are standby and suspend. When the PC goes into standby, the hard drive and monitors are turned off while everything else remains fully operational. The suspend mode completely shuts off the PC except for the CPU, which operates in minimal mode. Newer machines have the benefit of 'Suspend-to-RAM', which saves a snapshot of the current working environment to

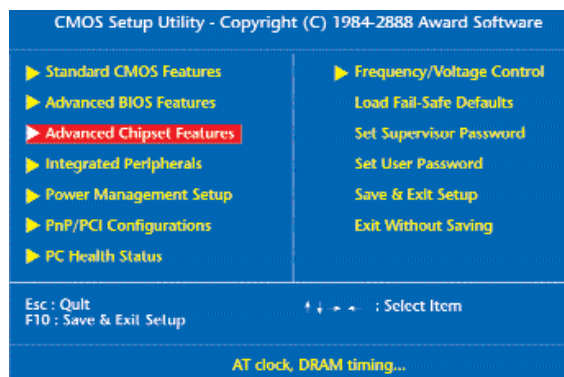
memory, which can be instantly recalled by a mouse click, or by hitting a key.

An easier way of managing these settings is through Windows itself. If your BIOS is ACPI-compliant, you can enable this option to easily configure through Windows as to how you want your computer to handle power saving issues.

■ **PC Health Status** essentially monitors the thermal condition of your PC. It basically shows a quick status of CPU and system temperature, voltage levels and fan speeds. Some computers come with a facility to set an alarm and turn off the PC in case the temperature exceeds a specified limit. In short, it's a simple way of making sure that nothing goes wrong, and nothing usually does.

■ **Passwords** lets you keep your computer as secure as possible by setting a password at a much lower level than the operating system. BIOSs generally impart two levels of security, one at the user level and the other at a system level.

A user-level password will let you boot up the computer as you normally would with restricted access to the BIOS. It is the system password that grants such authority to open and modify the BIOS. It's a perfect setup for PCs that have a supervisor with multiple users.



A typical view of the BIOS main menu



Commonly experienced problems, such as hard disk errors, can be easily fixed through standard BIOS options